

Prerequisites for Using K210 as a Coprocessor

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1. Import Model Files to Memory Card

Preface: Since AI vision experiments require different model files, it is necessary to import all model files to the memory card in advance. You can import all model files at once to avoid importing model files each time.

Model files are distinguished by different paths.

All model files are placed under the KPI folder. Please do not modify the file names in the KPI folder and subdirectories under KPI. Otherwise, the default example program will have the problem of not finding model files.

1. Prepare a card reader and a memory card (microSD card). The capacity should not exceed 64G. Insert the memory card into the card reader slot.



2. Connect the card reader to the computer USB port and open the root directory of the memory card drive. Here we take drive H: as an example.



3. Download the model file archive from the materials and extract it. Copy the KPI folder to the root directory of drive H:.

K210_SDCARD (H:)		
名称		修改日期
KPU		2023/1/11 15:17
main.py		2023/2/22 9:58

4. Eject the card reader and remove the memory card to insert into the K210 module's memory card slot.



2. K210 Auto-start AI Program

1. Insert the SD card into the card reader and then into the computer. Copy **one program** from the 11 programs shown in the figure below to the SD card.

image122

- 1_color_recognition_1.py
- 2_find_barcodes_2.py
- 3_find_qrcodes_3.py
- 4_find_apriltags_4.py
- 5_yolo_face_detect_ok_5.py
- 6_face_detect_68lm_6.py
- 7_face_mask_detect_7.py
- 8_face_recog_8.py
- 9_voc20_object_detect_9.py
- 10_self_learning_10.py
- 11_mnist_11.py

2. Rename it to a file named main.py.

U 盘 (F:) >		▼	🔄	在 U 盘 (F:) 中搜索	
名称	修改日期	类型		大小	
📁 KPU	2023/4/25/周二 16:49	文件夹			
📄 1_color_recognition_1.py	2023/5/25/周四 10:32	Python File		5 KB	

3. Remove the SD card from the card reader, insert it into the K210's SD card slot, then power on the K210.

After waiting for a period of time, the main.py program in the SD card will run, which means the AI program has started.

3. Notes

1. If you want to know what each program sends, you can check the communication protocol under the k210 folder in the program source code folder.
2. If you want to learn how the AI program performs image processing and recognition and displays it on the LCD screen, you can refer to Chapter 5 AI Program for program source code interpretation.
3. This chapter focuses on K210 as a coprocessor, focusing on how to communicate with the upper computer and transmit data to the upper computer. Therefore, the explanation of AI image processing programs will not be involved.